

Ioan Alexandru Popa

 ALEX11BR |  alex11br.github.io |  alexioanpopa11@gmail.com |  LinkedIn |  +40 765 081 127

EDUCATION

Aarhus University, **Master in Computer Science**

 Aug 2026 - Jun 2028

- Relevant Coursework: Compilation (**OCaml**), Formal software verification (**Rocq**)

Politehnica University of Bucharest, **Bachelor in Computer Science** (Engineering)

 Oct 2022 - Jul 2026

- Relevant Coursework: Programming (**C**, **Python**; **Assembly**; **Rust**), Data structures and algorithms, Object-oriented programming (**Java**), Operating systems, Databases (Oracle **SQL**), Cloud computing (**Kubernetes**)
- **Cumulative grades:** 9.22/10

WORK EXPERIENCE

Nvidia

 Helsinki, Finland

Software Infrastructure Engineering Intern

 Jun 2025 - Sep 2025

- Reduced by **99%** the duration of action comparisons by implementing an **HTML** endpoint that compares 2 actions in an internal **Go** backend.
- Automated **4** steps of an OS image validation process using **Python** and **Bazel**.

PROJECTS

Personal dotfiles management system ([GitHub link](#))

 Mar 2021 - May 2026

- Created a GitHub repository with configuration files for select programs, like **vim**, **zsh**, **VSCode**.
- Implemented **shell scripts** for **5** distro families that install a couple apps and configure a freshly installed **Linux** system with favourite configurations, and a **PowerShell** script that does this for **Windows**.

ThemeChanger ([GitHub link](#))

 Aug 2021 - May 2025

- Designed a Linux app in **Python** and **GTK3** that lets the user modify the mouse cursor, application icon, and widget themes and settings of **4** theme frameworks, even for unthembable libadwaita apps.
- Implemented live theme reloading using **6** desktop environment-specific mechanisms.
- Built a mechanism of showing instantly GTK3 theme and CSS changes in the app.

emscripten-functions ([GitHub link](#))

 Aug 2023 - Mar 2025

- Implemented raw **Rust** bindings for **emscripten** system functions. Emscripten is a compiler toolchain that allows C & C++ code to be ran on web pages using **WASM**.
- Built and documented Rust-friendly wrappers for **29** emscripten-specific functions.
- Over **12,000** downloads on [crates.io](#) for the [emscripten-functions](#) crate (the Rust-friendly wrappers), and over **14,000** for [emscripten-functions-sys](#) (the raw Rust bindings).

VOLUNTEER ACTIVITIES

Undergraduate teaching assistant, Assembly programming

 Feb 2024 - Jun 2024

- Held a weekly laboratory session for over **10** students that shows them the basics of **x86 Assembly**.
- Made a **Docker** image for lecture demos.

OPEN SOURCE CONTRIBUTIONS

• Added the ThemeChanger application to nixpkgs. (**Nix**)

 Oct 2021 - Feb 2025

• Enhanced the tests suite of Lemmy and rust-vmv/kvm-ioctls. (**Rust**)

 Jan 2024

• Fixed Romanian city names in OpenTTD. (**C++**)

 Mar 2022